Credit Risk Model

1. Introduction

Credit risk modeling is an essential application of machine learning in the banking and financial sector. Financial institutions use credit risk models to estimate the probability that a borrower will default on a loan. Accurate prediction of credit risk helps banks reduce financial losses, improve loan approval decisions, and manage overall portfolio risk.

The objective of this project is to build a machine learning model that predicts whether a loan applicant is likely to default based on demographic, financial, and credit-related information.

2. Objective

The primary objectives of this project are:

- Predict whether a borrower is likely to default on a loan.
- Analyze customer financial and credit information.
- Build a reliable classification model using Logistic Regression.
- Evaluate model performance using standard classification metrics.
- Deploy the trained model through a Streamlit web application.

3. Dataset Description

The dataset contains information about loan applicants and their financial history. It consists of multiple features that influence credit risk.

Features Used

- Person Age
- Person Income
- Home Ownership
- Employment Length
- Loan Intent
- Loan Grade
- Loan Amount
- Interest Rate
- Loan Percentage of Income
- Previous Loan Default History
- Credit History Length

Target Variable

- **loan_status**
  - 0 = No Default
  - 1 = Default

4. Methodology

The project was completed using the following workflow:

Step 1: Data Collection

The credit risk dataset was imported into a Jupyter Notebook using the Pandas library.

Step 2: Data Preprocessing

The preprocessing stage included:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Checking data types
- Preparing features for machine learning

Step 3: Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset, including:

- Loan Status Distribution
- Income Distribution
- Loan Amount Distribution
- Correlation Heatmap
- Feature Distribution Analysis

Step 4: Feature Engineering

Relevant features were prepared and transformed to improve model performance.

Step 5: Model Training

The dataset was divided into training and testing sets using an 80:20 ratio.

A Logistic Regression classifier was trained on the processed dataset.

Step 6: Model Evaluation

The trained model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix
- Classification Report

5. Tools and Technologies

The project was developed using the following technologies:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib
- Streamlit

6. Results

The Logistic Regression model successfully classified loan applicants into default and non-default categories based on their financial and credit information.

The evaluation metrics demonstrated that the model can effectively assist in identifying high-risk borrowers. The trained model was saved as `credit_risk_model.pkl` and integrated into a Streamlit web application for real-time prediction.

7. Applications

The developed credit risk model can be used in:

- Commercial Banks
- Financial Institutions
- Loan Approval Systems
- Credit Card Companies

- FinTech Companies
- NBFCs (Non-Banking Financial Companies)

8. Future Scope

The project can be further improved by:

- Implementing Random Forest and LightGBM models
- Hyperparameter tuning
- Cross-validation
- SHAP-based model explainability
- Automated feature engineering
- Cloud deployment using Streamlit Community Cloud, Render, or AWS
- Integration with real-time loan management systems

9. Conclusion

This project demonstrates the practical application of machine learning in predicting credit risk. By analyzing customer financial and credit information, the model assists lenders in making informed loan approval decisions while reducing the likelihood of financial loss. The complete workflow, from data preprocessing to deployment, provides a strong foundation for real-world credit risk assessment systems.